

Aryaman Sharma

Toronto | [linkedin.com/in/aryaman77/](https://www.linkedin.com/in/aryaman77/) | github.com/arymn7 | a765shar@uwaterloo.ca | 647-534-7002

EDUCATION

University of Waterloo - **STUDENT ID: (21192935)**

Waterloo, Canada

B.C.S. Computer Science Honours Co-op (GPA: 3.70/4.0)

September 2025 – April 2030 (Expected)

- **Awards:** President's Scholarship of Distinction, 1x Term Distinction
- **Coursework:** Elementary Algorithm Design and Data Abstraction, Designing Functional Programs, Calculus 1 & 2, Algebra with Proofs, Linear Algebra 1, Tools/Techniques for Software Development, Introduction to Microeconomics

PROJECTS

Heart Disease Prediction | Python, scikit-learn, NumPy, Pandas, Git **Link:** github.com/arymn7/heart_disease_prediction

- Developed an end-to-end **machine learning pipeline** to predict **heart disease risk**, achieving **~85% classification accuracy**, using **logistic regression** on **300+ patient records**.
- **Improved model reliability** through **data cleaning and feature engineering**, validated via an **80/20 train–test split**, using **Python, Pandas, NumPy, and scikit-learn**.
- Evaluated **model performance** on a held-out test set using **accuracy and classification metrics**, ensuring robust generalization, with **scikit-learn's** evaluation framework.

RC Vehicle | C++, Arduino, Embedded systems, Electronics

- Designed and built an **RC car** using **Arduino, C++ firmware** using **structured, modular code** and **interrupt-based logic**, achieving a top speed of **15 km/h** with **low-latency real-time motor control**.
- Integrated **embedded systems** components for **wireless operation**, delivering a **~30 m control range** with **stable signal performance** and reliable **system responsiveness**.

Car Marketplace | Java, Git

Link: github.com/arymn7/car-marketplace

- Developed a **terminal-based car marketplace** application, enabling **vehicle listings, search/filtering**, and **buy/sell workflows**, using **Java** and **object-oriented design** principles.
- Improved code quality by designing a **modular architecture** and collaborating via **Git** for **feature development, debugging, and code integration**, using **industry-standard version control** practices.

BlackJack | Python, Pygame, Git

Link: github.com/arymn7/blackjack-pygame

- Built a **fully interactive Blackjack** game with **real-time rendering** and **input handling**, supporting **10+ game states** and complete win/loss/draw resolution, using **Python** and **Pygame**.
- Implemented **robust game logic** with hit/stand actions, automated dealer behavior (hit until 17), score calculation, and bust detection, scaling gameplay to **200+ cards** with **randomized shuffling**, using **object-oriented design**.
- Designed a **user-friendly graphical interface** with **restart/replay functionality**, ensuring **smooth gameplay** and **maintainability**, while managing **iterative development** and **bug fixes** via **Git** with structured commits.

EXPERIENCE

Fletcher's Meadow Secondary School

Toronto, Canada

Peer Tutor - Computer Science & Mathematics

September 2023 – June 2025

- Delivered **150+ hours** of support in computer science, teaching **object-oriented programming** principles in (**Java/Python**) and guiding peers through coding projects, which **boosted coding proficiency** and **problem-solving**.
- **Tutored 15+ peers** in calculus by simplifying **complex concepts** like derivatives, integrals, and limits into accessible explanations, reinforcing students' strategies while leading to **20–30% improvements** in test performance.

SKILLS

Languages: C, C++, Python, Java, HTML, CSS, JavaScript, Racket, SQL

Technologies: Git, Pandas, Scikit Learn, NumPy, Pygame, React, NodeJS

Developer Tools: GitHub, VS Code, IntelliJ IDEA, PyCharm, Linux, Vim, Power BI, Excel

ACHIEVEMENTS

Certificate Of Distinction in the University Of Waterloo's Sir Issac Newton Exam (Physics)

May 2025

Certificate Of Distinction in the University Of Waterloo's Avogadro Exam (Chemistry)

May 2024